

# SOUVIK PODDAR

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Google Scholar: <https://scholar.google.com/citations?user=HyodwH8AAAAJ&hl=en&oi=ao>

Portfolio: <https://souvikpoddarphd.github.io/portfolio/>

## EDUCATION

### University at Buffalo

Doctor of Philosophy, Mechanical Engineering, GPA: 3.80/4.00

Buffalo, NY

May 2026

### Arizona State University

Master of Science, Mechanical Engineering, GPA: 3.72/4.00

Tempe, AZ

May 2020

### National Institute of Technology

Bachelor of Technology, Production Engineering, GPA: 8.34/10

Agartala, India

May 2018

## PEER-REVIEWED JOURNAL PAPERS

- **Poddar et al.** A Robotic Emulator for User-Driven Design of Multi-DOF Transradial Prostheses. IEEE Robotics and Automation Letters. 2026. DOI: [10.1109/LRA.2026.3664529](https://doi.org/10.1109/LRA.2026.3664529)
- **Poddar et al.** Improving Foot-Rocker via Robot-Resisted Gait Training with Self-awareness Biofeedback in Adults with Cerebral Palsy. IEEE Transactions on Neural Systems and Rehabilitation Engineering. 2025. DOI: [10.1109/TNSRE.2025.3636432](https://doi.org/10.1109/TNSRE.2025.3636432)
- **Poddar et al.** Understanding Weight and Performance Trade-offs in Transradial Prostheses using a Robotic Emulator. 2026. Manuscript under preparation.
- Sridar, S., **Poddar, S.**, Tong, Y., Polygerinos, P. and Zhang, W. Towards untethered soft pneumatic exosuits using low-volume inflatable actuator composites and a portable pneumatic source. IEEE Robotics and Automation Letters. 2020. DOI: [10.1109/LRA.2020.2986744](https://doi.org/10.1109/LRA.2020.2986744)
- Lee, J.\*, Kapahi, S.\*, **Poddar, S.** and Kang, J. A personalized enhanced feedforward force control detecting cyclic disturbance for gait rehabilitation robot. IEEE Transactions on Medical Robotics and Bionics. Manuscript in revision. 2024.
- Quinones Yumbla, E., Rokalaboina, J., Kanechika, A., **Poddar, S.**, Nibi, T.M. and Zhang, W. Gait Sensing and Haptic Feedback Using an Inflatable Soft Haptic Sensor. ASME Letters in Dynamic Systems and Control. 2024. DOI: <https://doi.org/10.1115/1.4064377>

## PEER-REVIEWED CONFERENCE PAPERS

- **Poddar et al.** A lightweight transradial prosthetic emulator for optimizing prosthetic wrist design. IEEE International Conference on Biomedical Robotics and Biomechatronics. 2022. DOI: [10.1109/BioRob52689.2022.9925286](https://doi.org/10.1109/BioRob52689.2022.9925286)
- **Poddar et al.** A cable-actuated prosthetic emulator for transradial amputees. Annual International Conference of the IEEE Engineering in Medicine & Biology Society. 2021. DOI: [10.1109/EMBC46164.2021.9631108](https://doi.org/10.1109/EMBC46164.2021.9631108)
- Amatya, S., Lafmejani, A.S., **Poddar, S.**, Sridar, S., Sugar, T. and Polygerinos, P. Design, development, and control of a fabric-based soft ankle module to mimic human ankle stiffness. IEEE International Conference on Rehabilitation Robotics. 2019. DOI: [10.1109/ICORR.2019.8779495](https://doi.org/10.1109/ICORR.2019.8779495)

## RESEARCH EXPERIENCE

### Research Assistant, University at Buffalo

Buffalo, NY (Aug 2020 - Present)

- Developed I-PEDLE (Intelligent Prosthesis Emulator for Daily Living Enhancement), a lightweight (380 g) 3-DOF robotic emulator for user-driven evaluation of multi-DOF transradial prostheses to inform prosthesis selection and prescription.
- Executed a human-subject study (n=10) with I-PEDLE to quantify how systematically varied wrist function affects task performance and upper-limb kinematics.
- Implemented robot-resisted gait training with biofeedback for adults with cerebral palsy (n=5), improving heel contact, posture, lower-limb strength, and spatiotemporal gait measures.

- Managed compliance and study operations across 3+ IRB-approved studies, including multi-site protocols, regulatory documentation, recruitment materials, SOPs, and session workflows.
- Mentored Master's and undergraduate students on research projects and deliverables.

#### **Visiting Scholar, University of Michigan**

**Ann Arbor, MI** (Jan 2025 - Jan 2026)

- Led an I-PEDLE user study (n=15), systematically varying prosthesis weight and quantifying effects on task performance, upper-limb kinematics, and perceived workload (NASA-TLX).
- Translated study results into design recommendations by identifying trade-offs between prosthesis weight, functional performance, user workload and fatigue.

#### **Research Assistant, Arizona State University**

**Tempe, AZ** (Aug 2018 - May 2020)

- Modeled and optimized inflatable actuator composites using FEA sweeps (33/66/100% volume) and benchtop validation. Achieved ~4.5x faster pressurization.
- Contributed to building a portable pneumatic source and control system (0.131 MPa max pressure, 21.45 SLPM max flow) for wearable-robot operation.
- Designed and characterized a fabric-based soft ankle module to emulate ankle stiffness. Reached ~50% of biological ankle stiffness.
- Developed inflatable insole sensors for gait-phase detection. Fabricated and validated load response up to 700 N.
- Automated Python-driven FEA simulations across actuator modes (stiffening, contracting, elongating). Validated via benchtop tests.

#### **TEACHING EXPERIENCE**

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- Teaching Assistant (Aug 2020 - May 2022): Road Vehicle Dynamics; Manufacturing Processes; Introduction to Mechanical & Aerospace Engineering Practice
- Supported course delivery for 200+ students by grading assignments and exams, proctoring classes, leading recitation and office hours, and assisting with hands-on course projects, including MATLAB code debugging and 3D printing support.

#### **PATENTS, AWARDS AND FELLOWSHIPS**

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- U.S. Patent US 12,458,556 B2 (Issued 2025): Low-volume inflatable actuator composites for wearable applications and portable pneumatic source.
- Research supported by the NSF Disability and Rehabilitation Engineering (DARE) program award (Aug 2022 - Present).
- Master's Opportunity for Research in Engineering (MORE) award (Aug 2019 - May 2020).

#### **POSTER PRESENTATIONS AND TALKS**

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- Gait Sensing using Fabric-based Soft Inflatable Actuators and Force Haptic Feedback. Poster Presentation. MORE Spring 2020.
- Towards Portability of Pneumatic Wearable Devices using Low-Volume Inflatable Actuator Composites and a Portable Pneumatic Source. Poster Presentation. MORE Fall 2019.
- A soft inflatable exosuit to assist knee swing during walking. Oral Presentation (on behalf of RISE Lab, Wenlong Zhang), ASU Southwest Robotics Symposium, Arizona State University. 2018.